Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2023	ENGI3371-WE01

Title: (BEng/MEng)

Electrical Engineering 3

Time Allowed:	2 hours
Additional Material provided:	No
Reporting problems during the exam:	Queries about the instructions for this exam (rubric) should be directed to: eng.boe@durham.ac.uk. If you experience any difficulties uploading your exam answers to Gradescope report these immediately to: eng.boe@durham.ac.uk.

Instructions to Candidates:	This is a synchronous, time restricted exam. All submissions must be uploaded to Gradescope as per the published timetable. Answer ALL questions. Please answer each question on a new page. All workings must be shown. This is an online open-book exam. • *The examination attempt must be all your own work. You must not interact with another person (either one-to-one, by telephone, on the internet, in chat rooms, on social media, via smart watches, etc.), or use cheat sites, etc. whilst completing the exam. You must not discuss any examination papers with friends, relatives, current or past students of this or any other University before the end of the exam submission window.* • *Candidates are expected to answer all questions in their own words. Should close paraphrasing, direct quotation be necessary candidates must acknowledge the source of any material used in their answers (this includes text, images, diagrams, charts, tables and/or graphs). Full referencing is not expected, but a brief indication of the source of the material (e.g. name of author) must be given.*

Revision:	

Section A

Question 1

A three-phase full wave converter, shown in Fig. Q 1 is used to control the speed of a separately-excited d.c. motor rated at 100 kW, 600 V, 2000 rpm. The converter is connected to a three-phase 400 V, 50 Hz supply.

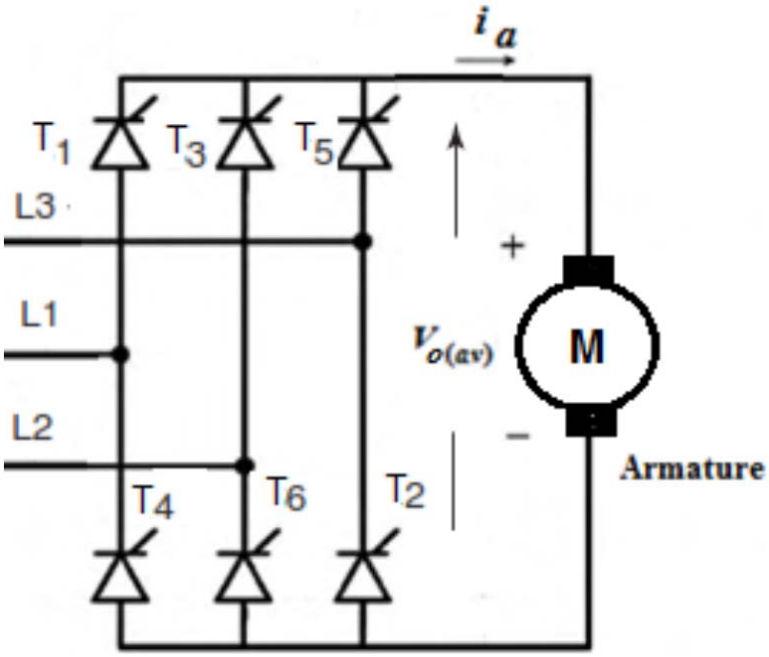


Figure Q 1 - three phase thyristor controlled DC motor

The armature resistance R_a = 0.051 Ω and the armature circuit inductance is L_a = 10 mH. The motor voltage constant is $K_e\Phi$ = 0.25 V/rpm. The rated armature current is 100 A and the no-load current is 10 A. The converter operates as a rectifier, and assume that the motor current is continuous and ripple-free.

(a) Draw the output waveform of the three-phase full wave thyristor converter supplying the armature winding and show that the average DC output voltage of the converter (i.e. the motor supply voltage) is determined by the following expression:

$$V_d = V_s \frac{3\sqrt{2}}{\pi} \times \cos\alpha$$

In which V_s is the RMS of the AC input single phase voltage and α is the thyristor converter firing angle. [40%]

(b) Determine:

(i) The no load speed when the firing angles are: α = 0º and α = 60º. [30%]

(ii) The firing angle to obtain the rated speed of 2000 rpm at rated motor current. [30%]

continued

Question 2

(a) A three-phase star connected induction motor is rated 35 HP, 400 V, four pole, 50 Hz. This motor drives an industrial blower load at rated voltage and frequency, and rated slip of 6 %. The torque-speed characteristics of the blower is given by:

$$T_m = 2.48 \times 10^{-3}(\omega_m)^2$$

Determine the speed and torque of the blower load when the motor is supplied by the rated voltage and rated frequency. [20 %]

(b) Sketch a block diagram of a doubly fed induction generator (DFIG) in a wind turbine and clearly label all components. Briefly explain the key role of the power electronic converter, and working principle of the system to maintain a constant frequency while wind speed fluctuates. [30 %]

(c) A three-phase star-connected 35 kVA, 400 V, 50 Hz, four pole synchronous machine has a synchronous reactance of $X_s = \mathrm{j}\,3\ \Omega$ per phase and negligible stator winding resistance. The machine is operating as a generator in parallel with the grid (infinite bus) at 100 % of the rating power, and 0.8 lagging power factor.

i) Determine the excitation voltage E_A and the phase angle δ between excitation voltage and terminal voltage. [30 %]

ii) Plot a phasor diagram including the terminal voltage, excitation voltage, armature current and voltage drop across the synchronous reactance. [20 %]

continued

Section B

Question 3

Consider a 50 Hz three-phase medium length transmission line with shunt admittances of $\frac{Y}{2} = 2e - 4\angle 90° \, S$ connected to the sending and receiving ends. For a receiving end phase voltage $127\angle 0$ kV the receiving end current is $I_R = 131.21\angle -36.9° \, A$, and the sending end phase voltage is $V_S = 130\angle 2°$ kV.

a) Find the generalised line constants (A, B, C, D).

[40%]

b) Find the magnitude of the receiving end voltage if the load current (at the receiving end) is $I_R = 100\angle -30° \, A$, and the sending end phase voltage magnitude is 127 kV.

[30%]

c) Find the magnitude of the receiving end voltage if the load at the receiving end is $25MW$ with a power factor of 0.9 lagging, and the sending end line-line voltage magnitude is 200 kV.

[30%]

continued

Question 4

a) Consider the short transmission line shown in Figure Q4 with the series reactance of $X = 0.3\,pu$, where a SSSC is inserted in series with the line and is injecting a voltage with magnitude of $|V_q| = 0.2\,pu$. The sending end and receiving end voltages are $V_s = 1.1\angle 15°\,pu$ and $V_r = 1.0\,pu$.

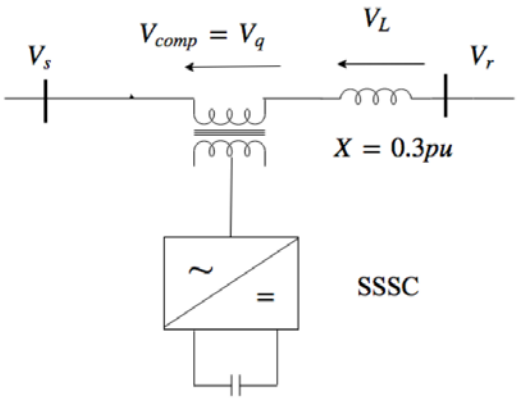


Figure Q4

Consider the two operating modes of the SSSC: (1) capacitive and (2) inductive modes. For each mode:

(i) Draw the phasor diagrams showing the relationship between V_s, V_r, V_q and the line current.
[30%]

(ii) Determine the reactive power delivered by the SSSC.
[20%]

(iii) determine the active power at the sending end of the line.
[20%]

b) In your own words, explain how FACTS technologies can be used to improve the operation of a power system.
[30%]

END